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# Contraceptive choices and preferences in a cohort of women with cystic fibrosis

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ABSTRACT

*Objectives:* As life expectancy for women with cystic fibrosis (CF) improves, reproductive decision-making is becoming increasingly relevant. We sought to characterize fertility intentions, contraceptive use, and preferences among adult females with CF.
*Study design:* We conducted a cross-sectional survey of contraceptive use and preferences in reproductive-aged women with CF. Subjects were recruited via an electronic mailing list from the Penn Adult Cystic Fibrosis Program.
*Results:* Fifty-three women with CF responded and met eligibility criteria. Most (83%) were sexually active, however only 49% used contraception. Condoms and the oral contraceptive pill were the most commonly used methods. Compared to the national contraceptive use among Caucasian women (65%), use in our cohort was significantly lower ($p = 0.01$). Participants prioritized effectiveness and ease of use in their contraceptive method; they wanted to avoid side effects and interference with sexual enjoyment. Although these priorities are similar to Caucasian women in the literature, additional contraceptive priorities differed significantly between the groups.
*Conclusions:* This CF cohort is sexually active but contraceptive utilization was low, and women with CF identify specific contraceptive preferences. There is a significant unmet need for contraception among women with CF.

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## 1. Introduction

Cystic fibrosis (CF) is one of the most common life-limiting inherited disorders, with an incidence of 1 in 3700 births in the United States [1]. Over the past several decades, medical advances have markedly improved the life expectancy of people living with CF. Whereas previously they were not expected to survive childhood, now the median age of survival is over 40 years, and half of the CF population is over 18 years old [2]. As more women with CF reach reproductive age, reproduction and family planning have become increasingly important.

Women with CF are sexually active [3], and most intend to have children [4–6]. However, one quarter of pregnancies are unplanned [5], and 15% result in abortion [7]. Although pregnancy is not thought to impact long-term survival [7], it can be associated with significant morbidity, such as infection, renal failure, and respiratory decompensation, and death can occur in patients with severe disease [8]. Therefore, contraception should play a central role in health care for women with CF so that unintended pregnancies can be minimized.

Hormonal contraception appears to be safe and efficacious in women with cystic fibrosis, and may even confer non-contraceptive benefits [9]. Research on contraceptive use in women with CF has been limited, and their contraceptive preferences are not well described. Previous studies estimate that 50–70% of women with CF use some form of contraception, with condoms and oral contraceptive pills being the most commonly used methods [4–6,10]. We sought to characterize current contraceptive choices and preferences of CF patients from a major hospital center in the United States.

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## 2. Methods

We conducted a cross-sectional web-based survey of reproductive aged women from 18 to 45 years old with CF via the electronic mailing list from the Penn Adult Cystic Fibrosis Program from 2014 to 2015. Patients who had received a lung transplant were excluded. Study data were collected and managed using REDCap electronic data capture tools.

Two published surveys were used to develop a single reproductive health and contraceptive preference instrument. An internally validated reproductive health questionnaire developed by Sawyer et al. [3], previously administered to women with CF, was used to ascertain reproductive health information. The *Contraceptive Attitudes Survey* developed by Foster et al. at the University of California San Francisco [11], a widely used survey designed for women at high risk of unintended pregnancy, was utilized to ascertain preferences since rigorously validated preference questionnaires specific to women with CF do not exist. By combining these surveys, we created a single reproductive health and contraceptive preference instrument specific to the reproductive-aged female CF population.

Development of the *Cystic Fibrosis Reproductive Health Survey* was overseen by an expert psychometrician. The survey underwent content and construct validity review with family planning experts and CF specialists as well as piloting for face validity by five female CF patients who completed the survey while waiting for care in the Penn Lung Center clinic waiting room. The final survey consists of 45 questions and takes approximately 20 min to complete.

Contraceptive use data from our CF cohort were compared to use among women across the United States according to the National Survey of Family Growth (NSFG), a nation-wide demographic survey of health and fertility conducted by the Centers for Disease Control and Prevention [12]. Results from the NSFG showed that contraceptive use differed significantly between Caucasian and African-American women. Since our CF cohort comprised Caucasian women almost exclusively, we controlled for the impact of race by limiting our comparisons to the Caucasian subgroup from the NSFG. A Pearson chi-square test was used to compare contraceptive use frequencies.

Bivariate analyses were performed to assess factors related to contraceptive choice and preferences. A generalized linear regression was used to estimate factors associated with contraceptive use and address potential confounders.

## 3. Results

Fifty-nine of 95 reproductive-age female CF patients participated, and 6 were excluded due to incompleteness of survey data, yielding a total of 53 participants. The mean age of the cohort was 30 and 94% were Caucasian (Table 1). Eighty-three percent of survey participants reported being sexually active. Twenty-one percent of participants reported a history of live birth, and 8% reported history of abortion. Demographic information for the non-responders was not available for comparison (see Table 2).

**Table 1**
Demographic data.

| | N = 53 |
|---|---|
| Age (mean) | 30 [range 17, 42] |
| Race | |
| Caucasian | 50 (94%) |
| Non-Caucasian | 3 (6%) |
| Sexually active | 44 (83%) |
| Pregnancy history | 14 (27%) |
| Live birth | 11 (21%) |
| Miscarriage | 3 (6%) |
| Abortion | 4 (8%) |

The oral contraceptive pill and condoms were the most commonly reported methods of contraception in this group of women with CF. Compared to a national sample of Caucasian women from the NSFG, this cohort of women with CF reported using monthly hormonal contraception (pills, patch, and ring) and highly-effective reversible contraception (implant and IUD) at similar rates (Fig. 1). However, within our cohort, condom use was twice as common as in the general population (17% vs 9%) and sterilization was much less so (2% vs 14%). Half of women with CF used no form of contraception at all, compared to 35% of the general Caucasian population ($p = 0.01$).

Women in this CF cohort prioritized minimization of side effects, effectiveness, a lack of interference with sexual enjoyment, and ease of use and access in a contraceptive method. Over half of women in this group found these characteristics to be extremely important.

Bivariate analyses of personal attributes and contraceptive use among our CF cohort initially revealed two variables with significant association: sexual activity and self-reported disease severity. Women with CF who were sexually active were 5 times more likely to use contraception as those who are not sexually active, and those with self-reported “severe” disease were half as likely to use contraception as those with self-reported “mild” disease. However, after a multivariable generalized linear regression to adjust for suspected confounding, neither association remained significant.

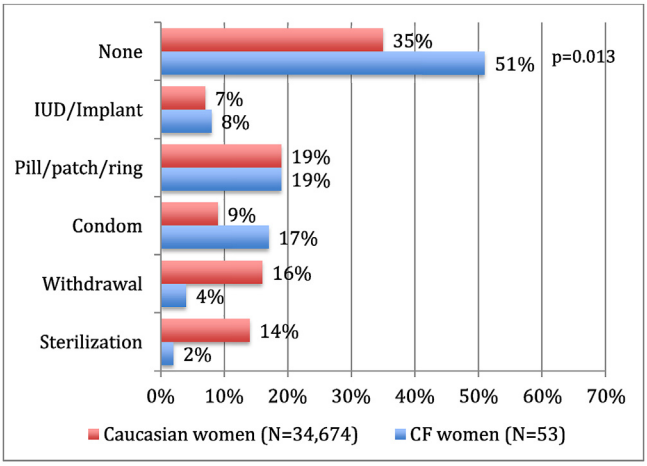
Other indicators of more severe disease, such as pancreatic insufficient genotype, low forced expiratory volume in 1 s, presence of diabetes, and recent hospital admission, were associated with lower contraceptive use (low body mass index was an exception to this trend). However, this was not statistically significant in our study population.

## 4. Conclusions

Despite high rates of sexual activity, this cohort of women with CF was less likely to use contraception compared to a general population of Caucasian women. This group sought effective, easy to use birth control that has minimal side effects and does not interfere with sexual enjoyment. Although many contraceptive

**Table 2**
Contraceptive characteristic preferences among women with cystic fibrosis.

| Contraceptive characteristic | Number of women who found the characteristic “extremely” important |
|---|---|
| Minimal side effects | 40 (77%) |
| Effective | 39 (75%) |
| Does not reduce enjoyment | 39 (75%) |
| Does not reduce partner's enjoyment | 37 (71%) |
| Easy to use | 37 (71%) |
| Easy to get | 32 (62%) |
| Affordable | 28 (54%) |
| Woman controls when and whether to use | 26 (50%) |
| Can be stopped any time | 25 (48%) |
| Do not need doctor to start | 22 (42%) |
| Woman responsible for use | 21 (40%) |
| Forgettable | 21 (40%) |
| Health benefit | 21 (40%) |
| Protects against STIs | 21 (40%) |
| Does not alter menstrual period | 21 (40%) |
| Undetectable | 20 (38%) |
| Immediate return to fertility after cessation | 18 (35%) |
| Used only at time of sex | 16 (31%) |




National Survey of Family Growth, 2011-2013.

**Fig. 1.** Contraceptive use among women with cystic fibrosis compared to a general population of Caucasian women.

methods fit these preferences, only half of this CF cohort used any form of contraception. These data suggest that women with CF may have an unmet need for contraception and have distinct contraceptive preferences.

In studies of women without CF, the top three contraceptive attributes were the same: effectiveness, minimal side effects, and affordability [11,13]. Although this cohort of women with CF shared many contraceptive preferences with the women studied elsewhere, they prioritized effectiveness, affordability, personal responsibility, forgettable use, and undetectable use less than their healthy counterparts.

Our study had a small sample size and was drawn from a single center, which may limit generalizability to the greater population of women with CF. However, our center is one of the largest in the United States and serves urban, suburban, and rural populations. The small sample also limited our ability to adjust for confounders that may have impacted contraceptive preferences and utilization. As of this writing, this report on contraceptive use is only the second of its kind in North America. For relatively rare diseases like CF, clinic-based research reaches an inherently small cohort. Access to the nation-wide CF registry would provide larger numbers, though detailed reproductive and contraceptive information is not currently captured in these databases.

Further study is needed to understand CF patients' motivations to use or to avoid contraception. Their poorer health may impact reproductive choices in various ways. Women may be opposed to using an additional medication. Long-acting reversible contraception may be useful for women with CF, but we do not know how acceptable they find these methods. Alternatively, reproductive-aged women with CF may be more likely to be attempting to conceive than their peers in the general population, and therefore more likely to eschew contraception. Finally, women with CF may not know how easily they can become pregnant or which methods of birth control they may safely use. Education regarding fertility and contraception, especially in the absence of formal guidelines for CF in the *United States Medical Eligibility Criteria for Contraceptive Use*, may be lacking on the part of patients and providers alike.

In this cohort, patients with markers of severe disease tended to not use contraception, although this trend was not statistically significant. Further research is needed to ascertain if this pattern would bear out on a larger scale. If so, the sickest CF patients, for whom pregnancy is most dangerous, may constitute a subgroup that may warrant special attention and intervention.

As we learn more about the reproductive goals of women with CF, pulmonologists and family planning specialists should be alert to their underutilization of contraception so that we can better meet this population's contraceptive needs. The current study can provide preliminary data in order to further study reproductive care in patients with CF.

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## Conflict of interest

None.

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